Chapter 8

Programming Fundamentals in Python for Engineering Technology

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8.1. Introduction

Learning basic computer programming skills is important for engineering and technology students in their early years of college education. It is believed that with the modular programming strategy and the core programming skills, the students will be able to develop computer code to solve most engineering problems.

At Queensborough Community College, the department of Engineering offers Dual/Joint Degree Programs of Computer Science and Information Security A.S. degree. This Dual/Joint Degree prepares students from programs that rely on computing and quantitative methods, especially in areas related to digital forensics and cybersecurity. All the 64 credits of Computer Science and Information Security A.S. degree is transferable to baccalaureate degree programs at John Jay College of Criminal Justice.

On the other hand, for Computer Engineering Technology degree, ET509 Programming for embedded systems and ET575 Introduction to C++ programming Design and Implementation are major degree courses. While Electronics Engineering Technology on ET509 is a major degree course, and ET575 is an elective course.

8.2. Why Python?

A computer program is a set of instructions that causes a computer to perform some kind of action. Computers use multiple languages to communicate known as programming languages. A programming language is simply a particular way to talk to a computer in a set of instructions that both humans and the computer can understand. **Python** is an interpreted, high-level, general-purpose programming language. Created by Guido van Rossum and first released in 1991, Python's design philosophy emphasizes code readability with its notable use of significant whitespace. Its language constructs and object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects.

Python programming is an excellent choice for technology students because of its simplicity, versatility, and widespread use across various tech industries. With a clear and readable syntax, Python allows beginners to focus on learning programming concepts without getting overwhelmed by complex code structures. It supports a wide range of applications, from web development and data analysis to artificial intelligence and cybersecurity, making it a valuable tool for students pursuing careers in technology. Additionally, Python's large community and vast library ecosystem provide strong support and resources, enabling students to experiment, solve problems, and build real-world projects more effectively.